

ECO 4660-8666, International Economic Development, Fall 2024

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Problem Set 1

due Sep 23 (before class)

Please limit your answer in 3 pages. I don't need the raw data that you used for your analysis.

1. Update the table on page "Comparison of GNP" in slide "Chp1_2.pdf". Please state clearly the data source that you used.
2. Run a growth accounting for the following countries for the period of 1990-2020: China, India, South Korea, and Thailand. You should follow the steps below.
 - a. Get data on capital stock (K_t), labor supply (L_t), real GDP (Y_t), where $t = 1990 - 2020$; You can use the data from World Bank Open Data.¹
 - b. Calculate the TFP for each country: $TFP_t = \frac{Y_t}{K_t^\alpha L_t^{1-\alpha}}$, where you can assume a constant $\alpha = 0.33$ for all countries.
 - c. Calculate the annual growth rate for the above variables.
 - d. Compute the contribution of each factors to GDP growth. For example: the contribution of capital accumulation to GDP growth at a particular year is given by: $\frac{\alpha g_{K_t}}{g_{Y_t}}$, where g_{Y_t} is the growth rate of GDP and g_{K_t} denotes the growth rate of capital stock.
 - e. Interpret the results that you obtained in step d.

¹<https://data.worldbank.org/> or data from FRED <https://fred.stlouisfed.org>